

Handover: the 2026-08-27 arc — review, lane entry, fitting preparation

Written 2026-08-27 for a successor session. Read `handover.md` and `handover_2026-08-26.md` first; this file covers only this arc. The arc was Jonas's instruction, verbatim in scope: review the previous two days' implementation and documentation and correct as needed, reporting issues found; then settle the continuous lane-entry form and fix the desired-speed staging; then prepare the parameter fitting, including answering whether the fitting is methodologically feasible "with so many parameters".

1 The review: what was checked and what it found

The previous arc's own §9 flagged two things to verify; both were run, and the review went beyond them. Findings, in decreasing order of consequence:

1.1 The B-versus-C "statistically solid" claim was overstated, and is now corrected. The paired-difference analysis behind results doc §4.3b-ii ($\theta_C - \theta_B = 0.061$, 95% HDI [0.037, 0.107], "same sign on every resample") had no committed script. Reproducing it revealed it had resampled the reference while holding **both synthetic sides fixed**, omitting the synthetic-side sampling variance — which is independent between conditions and does not cancel in the difference. The committed reproduction (`replication/causation/paired_difference.py`, output `out/summary_paired_difference_fullp_abn.md`) runs all three schemes: `ref-only` reproduces the old numbers; under the project's own population convention the result is mean 0.055, 95% HDI [-0.006, 0.110], $P(\theta_C > \theta_B) \approx 0.97$. The ordering is strongly supported — consistent in point estimates at every bin count, corroborated by the braking aggregate (B within 0.3%, C over-braking 35%), ~ 0.97 posterior probability — but not "every resample". Corrected in results doc §4.3b-ii and §8, handbook ch. 08, README, and a bracketed correction in `handover_2026-08-26.md` §3.6. The irony that the arc devoted to uncertainty accounting produced one more incomplete uncertainty accounting is noted here so a successor keeps checking.

1.2 The stored bin-sensitivity summary and the results doc quoted intervals from different resampling conventions, without saying so. The stored sweep predated the population default; the doc's $N = 20$ remark quoted a third, intermediate run whose HDI matched neither. The sweep is regenerated under the population convention (its header now states the convention), and the θ -bias remark in §4.4 is corrected: under cases resampling B's $N = 20$ HDI [0.347, 0.773] excludes the point estimate 0.275; under the population convention it is [0.263, 0.389] and does not.

1.3 The §4.3 hand-copied intervals check out. Every θ , Θ , HDI and aggregate cell in §4.3/4.3b/4.3c matches `summary_fullp_abn.md` exactly.

1.4 Code defects found and fixed (none affected published numbers, verified): `equivalence_test` truncated weights to the filtered length instead of applying the finite mask — misaligning every weight-value pair after a mid-array NaN; all published inputs were checked NaN-free, so this never fired; fixed with a regression test. The resample error message omitted "population". `cutin_norm_weight`'s in-lane and adjacent bands tiled the whole corridor, leaving the straddling category empty by construction — the time-dependent norm could never engage; fixed (still neutral by default). `bin_sensitivity.py`'s `--abnormal` flag could never be turned off.

1.5 Role assignment misbehaves outside cut-ins, as §9 suspected: on the overtake traces it picks the cyclist as ego; on the truck-lateral traces (no vehicle moves laterally) the choice is arbitrary and picks a car 30 m behind. Cut-in traces (car and truck) assign correctly; LTAP finds the right target but the cut-in x/y conventions are meaningless in crossing geometry. Documented as a prerequisite of the transfer test (`docs/czb_fitting_plan.md` §5).

1.6 Documentation staleness fixed: duplicate section number 4.4 in the results doc (bin sweep and counterfactual rule; renumbered 4.5/4.6 with cross-references); the `handover.md` §2 row that said " $\Theta = 0.05$ derived from an aggregate-error bound" (the adopted value is 0.089; 0.05 is what a 5% tolerance would give); stale test counts.

2 The continuous lane-entry form (settled) and the staging fix

The full argument is `docs/lane_entry_note.md`; handbook ch. 04's open-problem section now records the resolution (`{{R4}}`, dark purple). In one paragraph: the binary lane gates are replaced by the **lateral overlap fraction predicted at the moment of longitudinal closure** (the lateral offset projected forward at its closing rate over the longitudinal TTC, clamped at centering — Jonas's $\tau_{\text{lat-versus-TTC}}$ comparison expressed as an overlap), and the saturated

safety magnitude by the **counterfactual residual impact speed** $\sqrt{\max(0, v_{\text{react}}^2 - 2 a_{\text{max}} d_{\text{avail}})}$, which is zero exactly at the model's own boundary $a_{\text{req}} = -a_{\text{max}}$ and turns the cliff into a ramp. Both changes are parameter-free; both reduce exactly to the released forms in every geometry the released scenarios sustain; both are flag-gated in `src/aidriver/preferences.py` (`lane_entry_continuous`, `counterfactual_residual_severity`, defaults off) so every published number is untouched. The note argues the reading on which this is the closed-form of the expectation the closed loop already computes by rollout, not a new preference. The desired speed is staged from the clip in `cutin_params` — the released scenarios stage $v_{\text{ego_des}}$ the same way.

Validation (`replication/czb/cutin_field_check.py`, `figures/cutin_field_check.png`): pre-onset deficit exactly zero in every clip; deficit ordered by criticality at every truncation point C2–C6 for cars and trucks; ramps, not steps, within the response window. 19 new property tests (`tests/test_cutin.py`) including the released-limit identities and a continuity test that caught a genuinely discontinuous first design (recorded in the note as a discarded intermediate).

Two empirical findings worth carrying:

- **The mild truck cut-ins (TTC6–8) have zero dread-field deficit through most of the response window** — at those gaps the counterfactual crash is avoidable at $a_{\text{max}} = 8$, so the field is silent where humans plausibly still press. The comfort–dread distinction (allowed deceleration below capability) is therefore a necessity for the fitting, not a refinement.
- **Onset detection was 1–2 frames late** (2%-of-span threshold), which put genuinely moving frames on the pre-onset side of the C1 anchor; fixed to an absolute 0.03 m threshold. Found via the C1 covariate check, which is the kind of thing only the data pipeline surfaces.

3 Fitting preparation (done) and the methods answer

`docs/czb_fitting_plan.md` is the document; `src/comfortzone/czb_data.py` the pipeline; `replication/czb/pilot_surface_fit.py` the committed stage-0 fit.

The answer to "do we have optimal methods with so many parameters": the fear is misplaced, by design. Everything upstream of the boundary is frozen (released parameters; parameter-free structural changes; staging from the clip), the stimulus field is precomputed once per clip, and all 3 096 Random cut-in trials share 18 covariate values. What remains is a generalized linear mixed model — ~5–8 population parameters plus shrunk per-driver random effects — for which MAP + Laplace in CPU-torch is entirely adequate, with a synthetic-recovery harness as the first step. No simulator in the likelihood, no CMA-ES, no ABC.

Stage-0 feasibility, already run: a two-parameter probit on the running-max deficit reproduces the study's 18-cell response surface with **correlation 0.90, RMSE 0.125** — and its residual (too flat in time within a criticality level) is precisely the accumulator's signature, motivating stage 2 before it is fitted. The Button pipeline works too: 2 396 trials, press times on the time-since-onset axis, medians ordered by criticality (TTC4 0.36 s → TTC8 1.8 s), only 0.5% never press.

4 What changed on disk

New: `docs/lane_entry_note.md`, `docs/czb_fitting_plan.md`, `handover_2026-08-27.md` (each with Word/PDF), `src/comfortzone/czb_data.py`, `tests/test_cutin.py`, `replication/czb/{cutin_field_check, pilot_surface_fit}.py`, `replication/causation/paired_difference.py` (+ its summary and log in out/), `figures/cutin_field_check.png`. Modified: `src/aidriver/preferences.py` (flags + `lane_entry_weight` + `residual_delta_v`), `src/comfortzone/cutin.py` (staging, obs fields, onset fix, norm-weight fix), `src/equivalence/test.py` (weight mask, error message), `replication/causation/bin_sensitivity.py`, `tests/test_causation.py` (regression test), `docs/crash_causation_results.md` (§4.3b-ii rewritten, §4.4 corrected, renumbering), `handbook chs. 04/08 + build_handbook.py` ({{R4}} dark purple), `README.md`, `handover.md`, `handover_2026-08-26.md` (correction note), regenerated `out/summary_bin_sensitivity_fullp_abn.md`. Suite: 123 tests, all passing.

5 Where I would pick up

- 1 **Synthetic-recovery harness, then the stage-1 hierarchical fit** on the Random cut-in surface (fitting plan §2–3). Jonas's four questions in plan §6 gate stage 1 but not the harness.

- ² **Add the truck cut-in cells** and run the deficit-versus-allowed-deceleration check (plan §4) — the place the data may overrule the scalar-level framing.
- ³ **The accumulator layer**, fitted on Random, validated on Button press-time densities with one paradigm shift.
- ⁴ **Transfer-scenario field constructions** — LTAP crossing geometry, role assignment by instructed-vehicle for overtake/truck-lateral (finding 1.5).
- ⁵ Standing items unchanged: Jonas's handbook review (now R1–R4), the Jian Wu email, the $a_{f,min}$ execution model, the epistemic gaze half.

6 Caveats

Written by the session that did the work. The stage-0 fit is population-level and deliberately naive; do not quote its c or σ as boundary estimates. The C1 covariate cells are small but nonzero for TTC6/8 (the anticipation weight activates on the first frame of motion); the design's bias term absorbs this, but anyone comparing C1 predictions to the "pure baseline" reading should know it is not exactly pure. The Button-side covariates at the press assume the clip's field applies up to the press frame; the 23 overshoot trials are dropped by the loader, per the dataset's own guidance.